

Question description

You are given an array A of N integers. Now, two functions $F(X)$ and $G(X)$ are defined:

$F(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] < A[Z]$

$G(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] > A[Z]$

Now, you need to find for each index i of this array $G(F(i))$, where $1 \leq i \leq N$. If such a number does not exist, for particular index i , output 1 as its answer. If such a number does exist, output $A[G(F(i))]$

Constraints:

$$1 \leq N \leq 3000$$

$$0 \leq A[i] \leq 10^{18}$$

Input :

The first line contains a single integer N denoting the size of array A . Each of the next N lines contains a single integer, where the integer on the i^{th} line denotes $A[i]$.

Output :

Print N space-separated integers on a single line, where the i^{th} integer denotes $A[G(F(i))]$ or 1 , if $G(F(i))$ does not exist.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

8
3 7 1 7 8 4 5 2

EXPECTED OUTPUT

1 4 4 4 -1 2 -1 -1

Test Case 2

INPUT (STDIN)

8
31 71 11 17 81 14 51 21

EXPECTED OUTPUT

11 14 14 14 -1 21 -1 -1

▼ Mandatory Test Cases

Test Case 1

KEYWORD

if(a[j]>a[i])

Test Case 2

KEYWORD

for(i=0;i<n;i++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

9

Test Case 2

TOKEN COUNT

225

Test Case 3

NLOC

30